



WeatherPost Systematic Workflow & Operations Manual

Multi-City Weather Content Automation Platform

Version 1.0 — Initial Release

Date June 1, 2026

Owner Bruce Love

Developer Lovable AI

Status Active Operations

Orlando, FL

Gainesville, FL

Miami (July 2026)

Scalable to 10+ Cities

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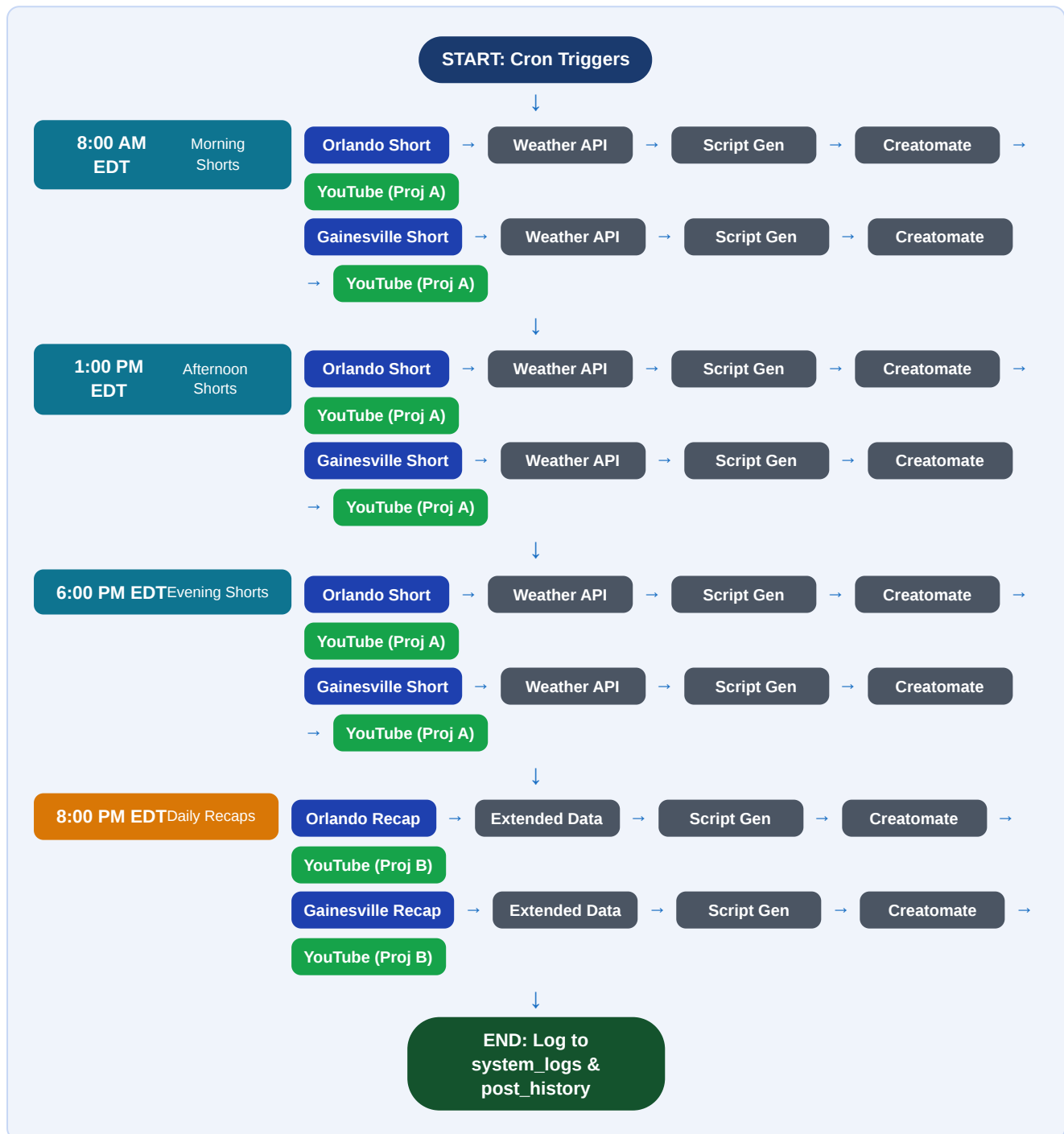
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SECTION 1 Daily Operations Workflow

Automated scheduling, success criteria, and nightly monitoring

1.1 Automated Daily Content Pipeline

The WeatherPost platform runs on a fully automated cron-triggered pipeline. Each day, six Shorts (three per city) and two Daily Recaps are generated, rendered via Creatomate, and uploaded to YouTube — all without manual intervention. The pipeline below shows the complete end-to-end flow for a standard operations day.



1.2 Success Criteria Checklist (Daily)

Every day should conclude with all items below verified as complete. Any unchecked item triggers the Bug Resolution Workflow (Section 4).

- ☒ All 6 Shorts posted (3 per city — Morning, Afternoon, Evening)
- ☒ All 2 Daily Recaps posted (Orlando + Gainesville at 8 PM EDT)
- ☒ All 8 posts recorded in `post_history` with `status = 'success'`
- ☒ All events logged in `system_logs` with no critical errors
- ☒ YouTube quota usage logged per project (Project A + Project B)
- ☒ Creatomate credit usage recorded for billing visibility
- ☒ Zero critical error notifications in SkyBrief notification panel

1.3 Daily Monitoring Checklist

Perform this review at **9:00 PM EDT** — after all scheduled content has been posted. This ensures issues are caught same-day and addressed before the next morning's cycle begins.

Checks to Perform

- ☐ Check SkyBrief notifications panel for any alerts
- ☐ Review `post_history` for today's posts (count = 8)
- ☐ Verify YouTube channels show new videos (both cities)
- ☐ Check `system_logs` for any error-level entries
- ☐ Review quota usage (Project A: ~9,600 units, Proj B: ~3,200 units)
- ☐ Review Creatomate credit usage vs daily target
- ☐ If any failures detected → Trigger Bug Resolution Workflow (Section 4)

Expected Daily Usage

Resource	Expected	Limit
Project A Quota	~9,600 units	10,000/day
Project B Quota	~3,200 units	10,000/day
Shorts Credits	~27.6 credits	≤5 ea.
Recap Credits	~70 credits	≤35 ea.
Total Credits/Day	~97.6 credits	Budget

SECTION 2 Weekly Operations Workflow

Sunday recap triggers, weekly review checklist, and Growth Lab check-in

2.1 Sunday Workflow (Weekly Recap Day)

Every Sunday, the standard 6 Shorts + 2 Daily Recaps pipeline runs as normal. Additionally, at **2:00 PM EDT**, the Weekly Recap process triggers automatically to generate and upload a comprehensive 7-day summary for each city.



2.2 Weekly Review Checklist

Complete this review every **Sunday at 9:00 PM EDT** after all weekly content has posted.

- | | |
|--|--|
| ☐ Verify both weekly recaps posted successfully | ☐ Document any issues encountered during the week |
| ☐ Review week's content performance (YouTube Analytics) | ☐ Update bug tracker if any open issues exist |
| ☐ Check Growth Lab experiment results for the week | ☐ Project B quota on Sunday: verify < 7,000 units |
| ☐ Review system uptime — scheduled posts success rate | ☐ Flag any anomalies for Monday morning review |

Sunday Quota Note

On Sundays, Project B processes both Daily Recaps (~3,200 units) AND Weekly Recaps (~3,200 units), totaling ~6,400 units. This is well within the 10,000 unit daily limit, leaving a healthy 36% buffer for retries or extra operations.

SECTION 3 Monthly Operations Workflow

Month-end recap triggers, monthly review, and strategic planning

3.1 Month-End Workflow

On the last day of each month, the Monthly Recap triggers automatically at **8:00 PM EDT** as an extension of the regular Daily Recap cycle. The monthly recap is the most comprehensive content type, featuring a full 30-day weather analysis.



Month-End Quota Note

On the last day of a non-Sunday month, Project B handles Daily Recaps + Monthly Recaps only (~6,400 units). On a Sunday month-end, all three types run: Daily + Weekly + Monthly (~9,600 units) — still within the 10,000

unit limit, but leaving only 400 units buffer. Monitor closely on these days.

3.2 Monthly Review Checklist

Complete this review on the **first day of the new month** to assess the prior month and set direction for the next.

Validation & Performance

- ☐ Verify both monthly recaps posted successfully
- ☐ Review month's content performance metrics (YouTube Analytics)
- ☐ Calculate actual credit usage vs budget target (3,882/month)
- ☐ Calculate actual quota usage vs project limits

Strategy & Planning

- ☐ Review Growth Lab wins/losses for the month
- ☐ Update Strategic Direction document if needed
- ☐ Plan next month's Growth Lab experiments
- ☐ Financial review: actual costs vs growth vs projections

SECTION 4 Bug Resolution Workflow

Severity classification, resolution process, and documentation templates

4.1 Bug Severity Classification

P0 — CRITICAL | Fix Immediately

- Content not posting (zero posts for any time slot)
- YouTube API errors preventing all uploads
- Creatomate rendering failures (all videos)
- Database connection failures
- Authentication / OAuth failures

P1 — HIGH | Fix Within 24 Hours

- Partial content posting (some succeed, some fail)
- Wrong metadata (incorrect month, city, title)
- Analytics not logging (post_history missing entries)
- Quota warnings (approaching 80% of daily limit)

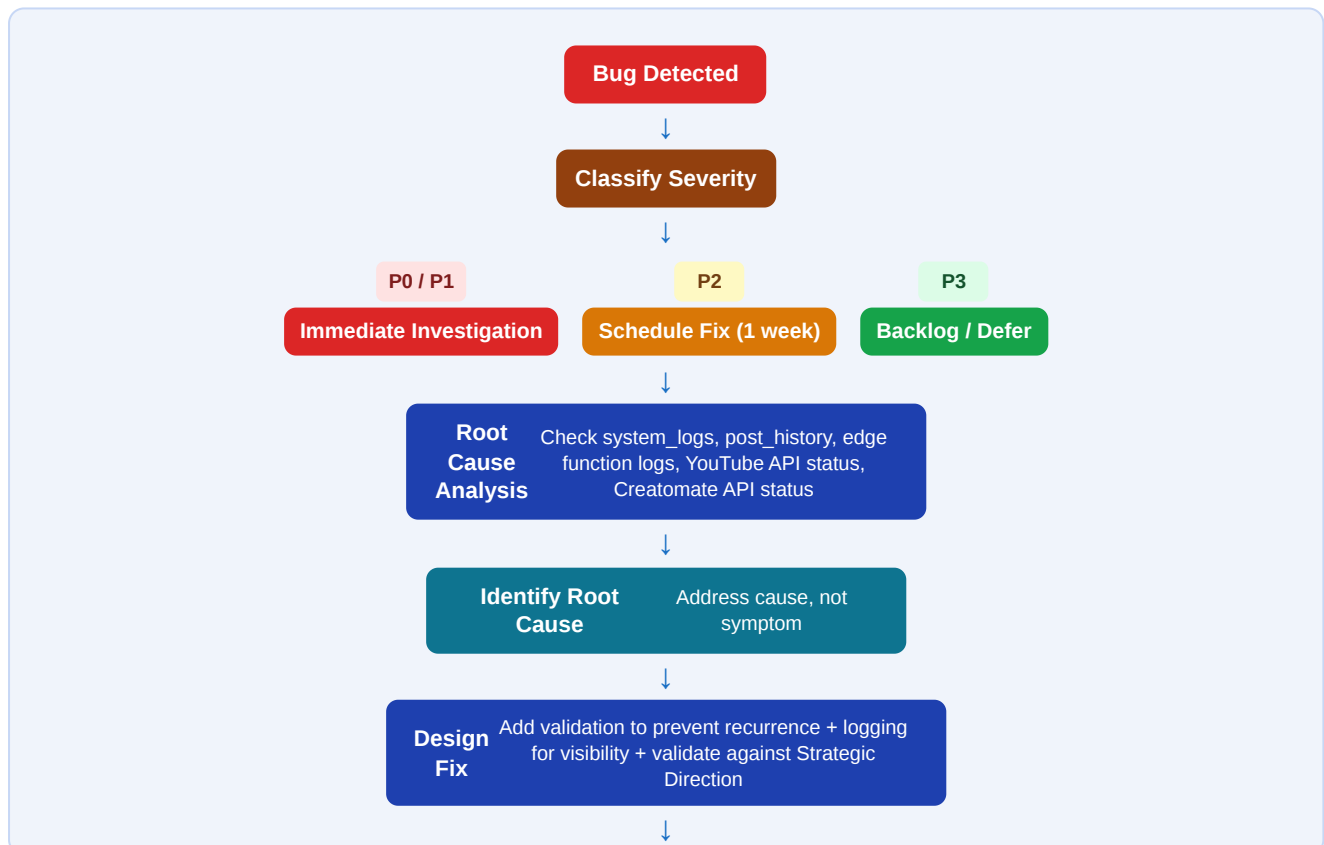
P2 — MEDIUM | Fix Within 1 Week

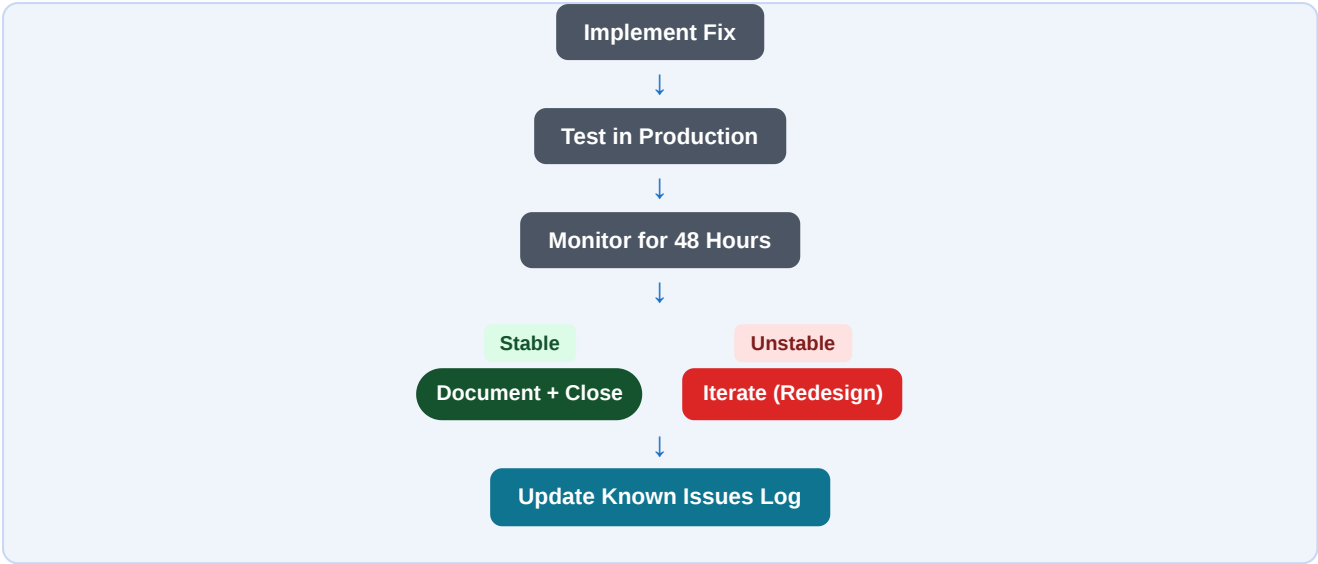
- Performance degradation (slow renders)
- Credit cost higher than target per video
- Minor UI issues in SkyBrief dashboard
- Non-critical logging gaps or missing fields

P3 — LOW | Fix When Convenient

- Cosmetic issues (visual inconsistencies)
- Nice-to-have feature requests
- Documentation updates or corrections
- Minor optimizations without impact

4.2 Bug Resolution Process





4.3 Bug Documentation Template

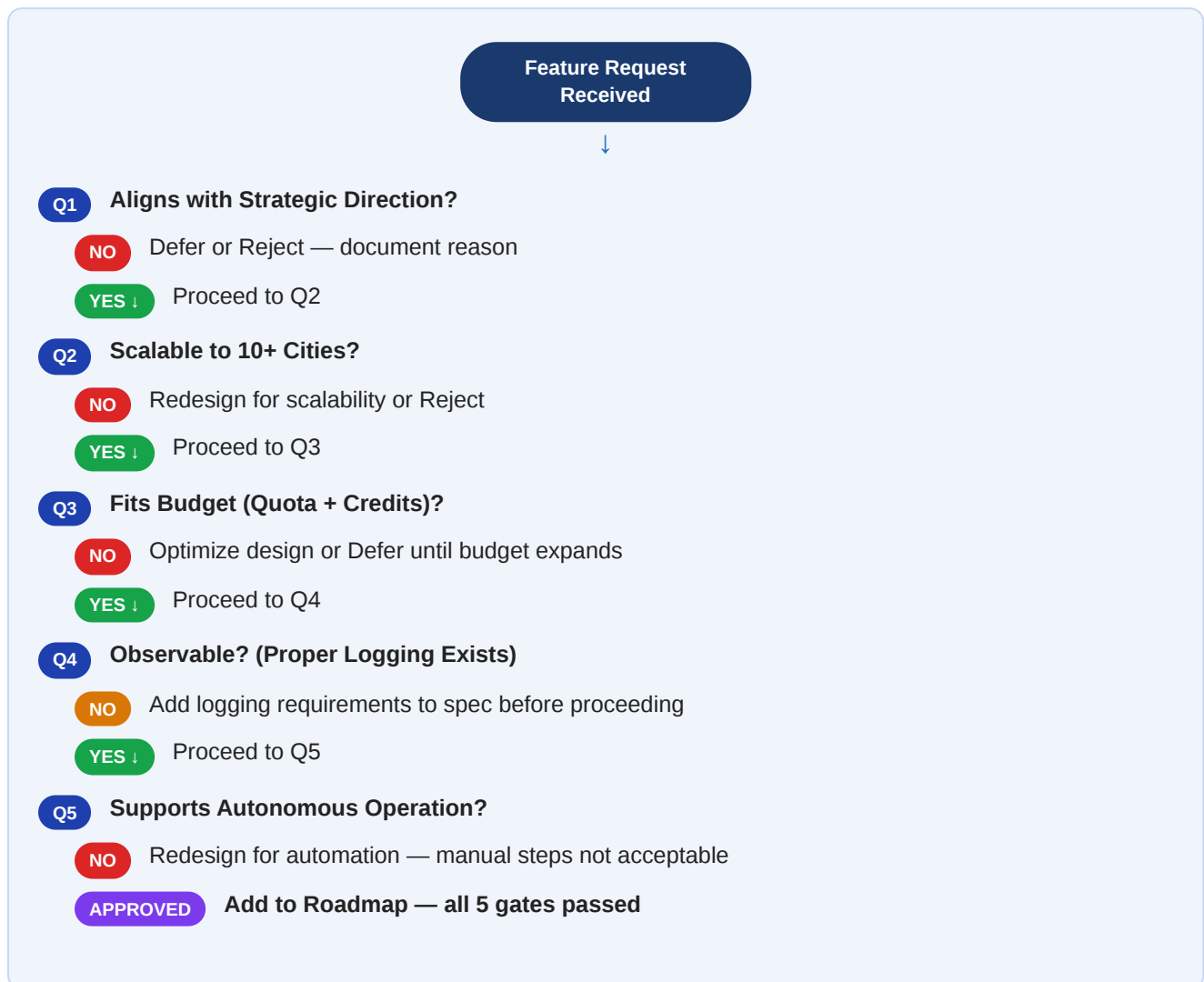
Field	Description	Example
Bug ID	Auto-increment identifier	BUG-001
Severity	P0 / P1 / P2 / P3	P1 — High
Discovered	Date and time of detection	2026-06-01 20:15 EDT
Symptoms	Observed failure behavior	"Orlando 8PM recap not posted"
Root Cause	Underlying technical cause	"OAuth token expired for Proj B"
Fix Applied	Change implemented	"Refreshed OAuth token, added auto-refresh logic"
Prevention	Safeguard to avoid recurrence	"Added 30-day token expiry alert"
Status	Open / In Progress / Resolved / Closed	Closed

SECTION 5 New Feature Implementation Workflow

Evaluation framework, approval gates, and 4-phase development process

5.1 Feature Request Evaluation Decision Tree

All feature requests — whether from the user, Growth Lab data, or Lovable AI's own observations — must pass through this five-gate evaluation before being approved for development.



5.2 Implementation Process (4 Phases)

Phase 1: Design

- 1 Write technical specification
- 2 Identify affected components
- 3 Plan database schema changes (if any)
- 4 Design logging strategy
- 5 Estimate credit and quota impact
- 6 **Get user approval before proceeding**

Phase 2: Development

- 1 Implement core feature logic
- 2 Add comprehensive logging throughout
- 3 Add error handling and recovery paths
- 4 Add input validation and guards
- 5 Write tests where applicable

Phase 3: Testing & Rollout

- 1 Test with 1 city first (canary release)
- 2 Monitor for 24–48 hours in production
- 3 Check logs for errors or anomalies
- 4 Validate actual budget impact vs estimate
- 5 If stable → scale to all cities

Phase 4: Documentation

- 1 Update Strategic Direction (if needed)
- 2 Update this Workflow Manual
- 3 Add entry to Known Features log
- 4 Notify user of successful completion

SECTION 6 Growth Lab Optimization Workflow

Experiment lifecycle, A/B testing protocol, and learning application

6.1 Experiment Lifecycle

Week 1: Design Experiment

Define hypothesis (e.g., "Question titles get higher CTR than statement titles") • Define metrics: CTR, watch time, subscriber growth • Define variants: A = control, B = test variant • Set minimum sample size (7+ days, 14+ posts for statistical significance)

Weeks 2–3: Run Experiment

Randomly assign posts to variants • Track performance metrics automatically via post_history • Log all engagement data (views, CTR, watch time) • Monitor for statistical significance as data accumulates

Week 4: Analyze Results

Compare variant performance side-by-side • Calculate confidence intervals • Identify clear winner (if any) • If inconclusive → extend experiment by 2 more weeks

Week 5: Apply Learnings

Promote winner to default for all future content • Deprecate losing variant • Apply winning approach to related content types • Roll out to all active cities simultaneously • Document findings in Growth Lab records

Ongoing: Iterate

Use learnings to inspire next experiment hypothesis • Build on wins cumulatively • Target 1–2 active experiments at any given time

6.2 Experiment Tracking Template

Field	Description
Experiment ID	Auto-increment (EXP-001, EXP-002...)
Content Type	Short / Daily Recap / Weekly Recap / Monthly Recap
Hypothesis	What specific outcome are we testing?
Variant A	Control — current default behavior
Variant B	Test — the proposed change
Metrics	Primary (CTR) + Secondary (watch time, subs)
Start Date	When the experiment began
End Date	Planned or actual end date
Sample Size	Number of posts in each variant

Results	Quantitative performance comparison (Variant A vs B)
Winner	A / B / Inconclusive
Action Taken	What changed as a result of this experiment?

SECTION 7

City Launch Workflow

Pre-launch checklist, launch day procedures, and city expansion timeline

7.1

Pre-Launch Checklist (New City)

2 Weeks Before Launch

- ☐ Existing cities stable (30+ days, zero P0 bugs)
- ☐ Credit budget has 25%+ unused buffer
- ☐ YouTube quota has 15%+ buffer per project
- ☐ Identify Growth Lab winners to apply to new city
- ☐ Create YouTube channel for new city
- ☐ Authenticate channel with appropriate OAuth project

1 Week Before Launch

- ☐ Configure city in SkyBrief settings
- ☐ Set up weather data sources for new city
- ☐ Test manual content generation (1 short + 1 recap)
- ☐ Verify videos render correctly in Creatomate
- ☐ Verify test video uploads to YouTube successfully
- ☐ Set automation schedules (shorts + recaps)

Launch Day

- ☐ Enable automation for all content types
- ☐ Monitor first 24 hours closely (manual oversight)
- ☐ Verify all scheduled posts succeed
- ☐ Check logs for any errors
- ☐ Validate quota usage matches expectations
- ☐ Validate credit usage matches expectations

Week 1 Post-Launch

- ☐ Daily monitoring (heightened vigilance)
- ☐ Compare new city performance vs existing cities
- ☐ Document any city-specific adjustments needed
- ☐ Confirm Growth Lab experiments include new city

7.2

City Launch Timeline

City 1 — Orlando, FL ✓ Active Launched: May 21, 2026 Status: Foundation city, generating optimization data Channel: Project A (Shorts) + Project B (Recaps) Advantage: First-mover, all Growth Lab data originates here
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Cities 5–10 — Florida Expansion Year 2

Tampa, Tallahassee, Fort Lauderdale, Pensacola, and additional Florida markets.

Target: Year 2 (2027–2028)

Advantage: Proven playbook, faster path to monetization per city. Each launch benefits from all prior Growth Lab learnings.

SECTION 8

Quota Management Workflow

Daily monitoring, weekly reviews, and escalation processes for YouTube API quota

8.1

YouTube Quota Monitoring

Project A — Daily Quota (Shorts)

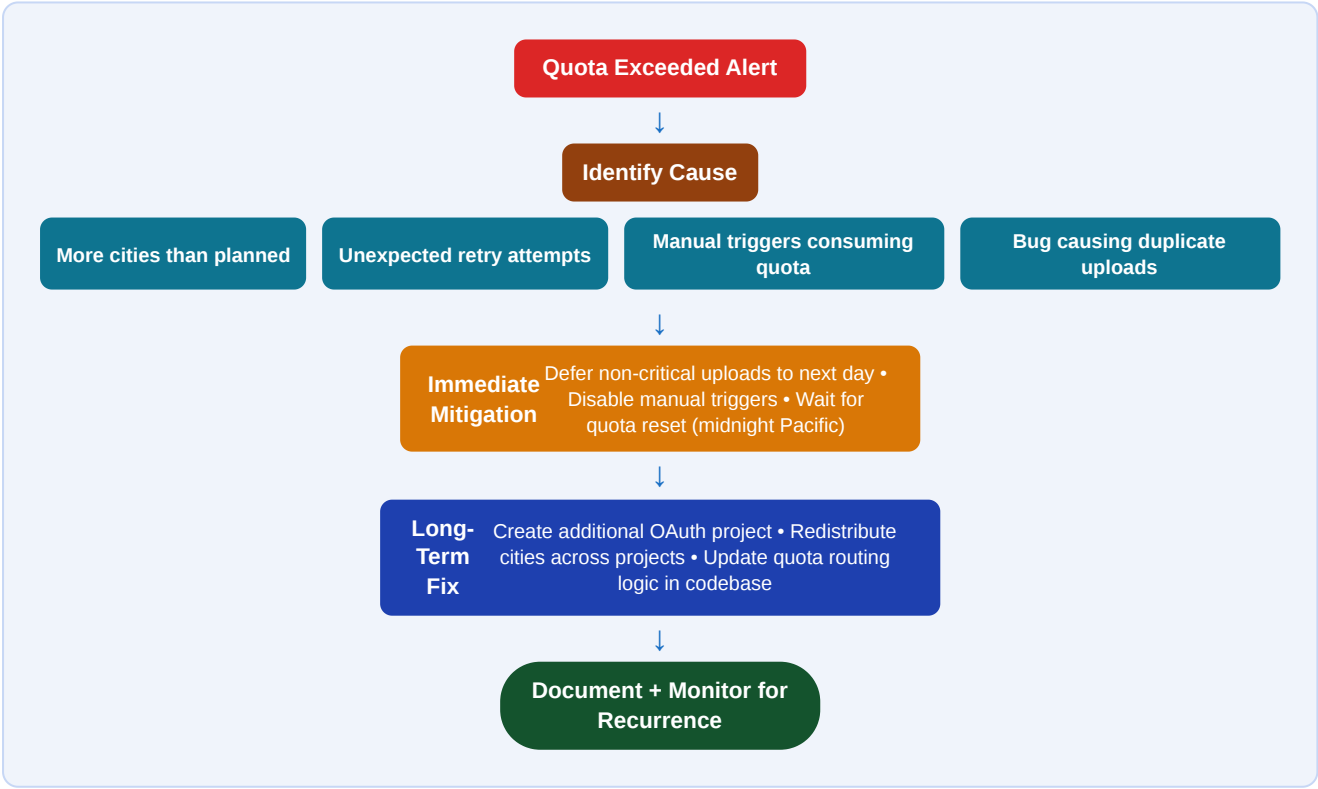
Scenario	Expected Units	Action
Normal (6 Shorts)	~9,600	✓ No action
Below 9,000	< 9,000	⚠ Investigate missing posts
Warning threshold	> 9,800	⚠ Monitor closely
Quota exceeded	> 10,000	🔴 Immediate alert

Project B — Daily Quota (Recaps)

Day Type	Expected Units	Buffer
Normal weekday	~3,200	68% remaining
Sunday (+ weekly)	~6,400	36% remaining
Month-end	~6,400	36% remaining
Sunday + month-end	~9,600	4% remaining

8.2

Quota Escalation Process



Weekly Quota Review Checklist

- ☐ Calculate average daily usage per OAuth project (7-day rolling average)
- ☐ Identify trends — is usage increasing or decreasing over time?
- ☐ Project future usage with planned city launches (3-city, 4-city scenarios)
- ☐ Plan new OAuth projects proactively — begin setup 3–4 weeks before needed

SECTION 9

Credit Budget Management Workflow

Monthly credit budget, alert thresholds, and optimization processes

9.1 Monthly Credit Budget Tracking

Current tier: **Growth (10,000 Credits/Month)**. The table below shows the planned credit allocation for 2 active cities.

Content Type	Posts/Month	Avg. Credits	Monthly Credits	% of Tier
Daily Shorts (2 cities × 6/day × 30)	360 posts	4.6	~1,656	16.6%
Daily Recaps (2 cities × 30)	60 posts	35	~2,100	21.0%
Weekly Recaps (2 cities × 4/month)	8 posts	72	~576	5.8%
Monthly Recaps (2 cities × 1/month)	2 posts	114	~228	2.3%
Growth Lab Experiments	Variable	Various	~150	1.5%
Total Target Usage	430 posts	—	~4,710	47.1%
Available Buffer (Safety Margin)			~5,290	52.9%

9.2 Credit Monitoring Alert Thresholds

50%

NORMAL

5,000 credits used. Operating as expected.

70%

MONITOR

7,000 credits. Watch closely, reduce experiments.

85%

INVESTIGATE

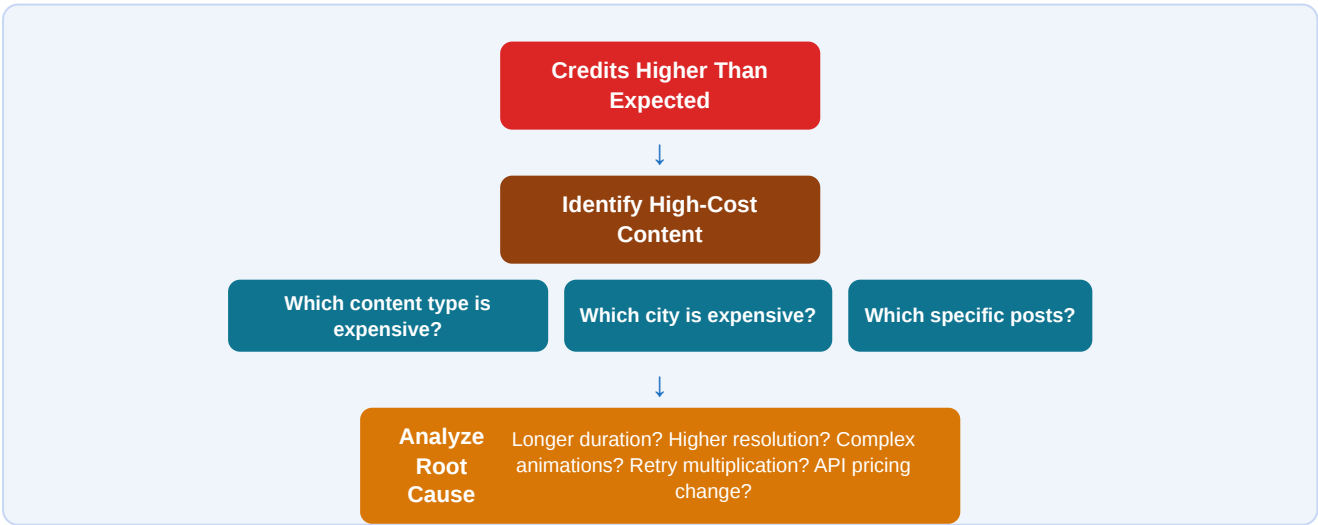
8,500 credits. Identify high-cost outliers immediately.

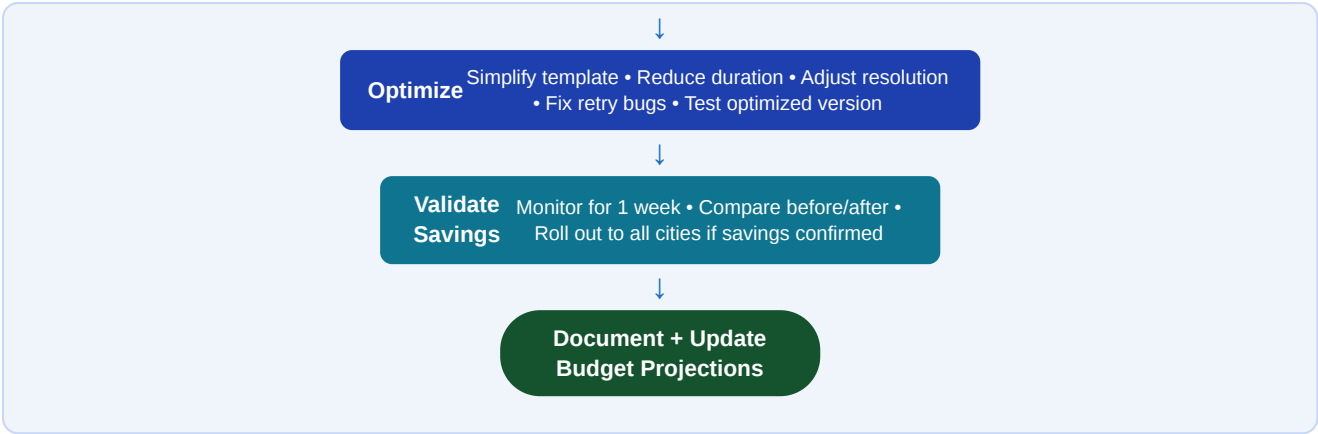
95%

URGENT

9,500 credits. Reduce consumption or upgrade tier now.

9.3 Credit Optimization Process

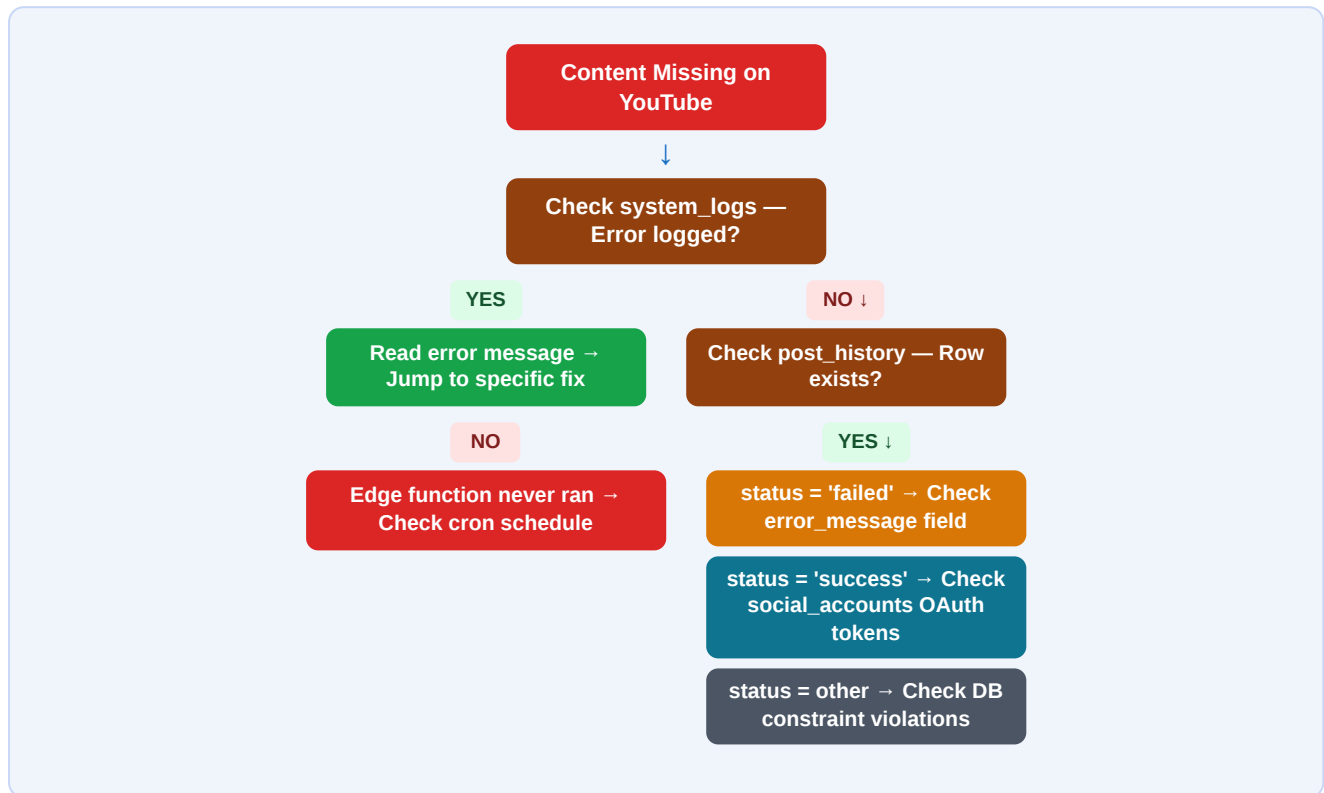




SECTION 10 Decision Trees

Operational decision guides for common scenarios and threshold decisions

10.1 Content Not Posting — Troubleshooting



10.2 Should I Create a New OAuth Project?

STEP Calculate current average daily units used (7-day rolling average)

- < 8K** No new project needed — healthy buffer remains
- 8K–9.5K** Plan new OAuth project now — set up within 2–3 weeks before adding new city
- 9.5K+** Create new OAuth project immediately — uploads at risk of failing daily
- > 10K** URGENT — Uploads already failing. Emergency project creation required today.

10.3 Should I Upgrade Creatomate Tier?

STEP Calculate average monthly credits used (trailing 3-month average for accuracy)

- < 1,500** Essential tier (2K credits) is sufficient — no upgrade needed
- 1,500–8,000** Growth tier (10K credits) recommended — current tier
- 8,000+** Premium tier (40K credits) needed — upgrade before hitting cap

10.4 Should I Add a New City?

Q1 Previous city stable for 30+ days with zero P0/P1 bugs?

NO Fix existing stability issues first — new city will amplify all current problems

YES ↓ Proceed to Q2

Q2 Credit buffer available? (25%+ unused monthly credits)

NO Optimize current credit usage or upgrade Creatomate tier first

YES ↓ Proceed to Q3

Q3 YouTube quota buffer available? (15%+ unused per project)

NO Create additional OAuth project and distribute cities across projects first

YES ↓ Proceed to Q4

Q4 Growth Lab showing improvements? (at least 2 experiments completed)

NO Wait for more Growth Lab data — launch with proven optimizations

YES ↓ Proceed to Q5

Q5 Monitoring capacity available for launch week?

NO Defer launch — heightened monitoring is required in first 7 days

ALL 5 GATES PASSED **PROCEED WITH CITY LAUNCH — follow Section 7 pre-launch checklist**

SECTION 11

Escalation Paths

When to contact the user vs. autonomous decisions vs. flag-only notifications

11.1

Decision Authority Matrix

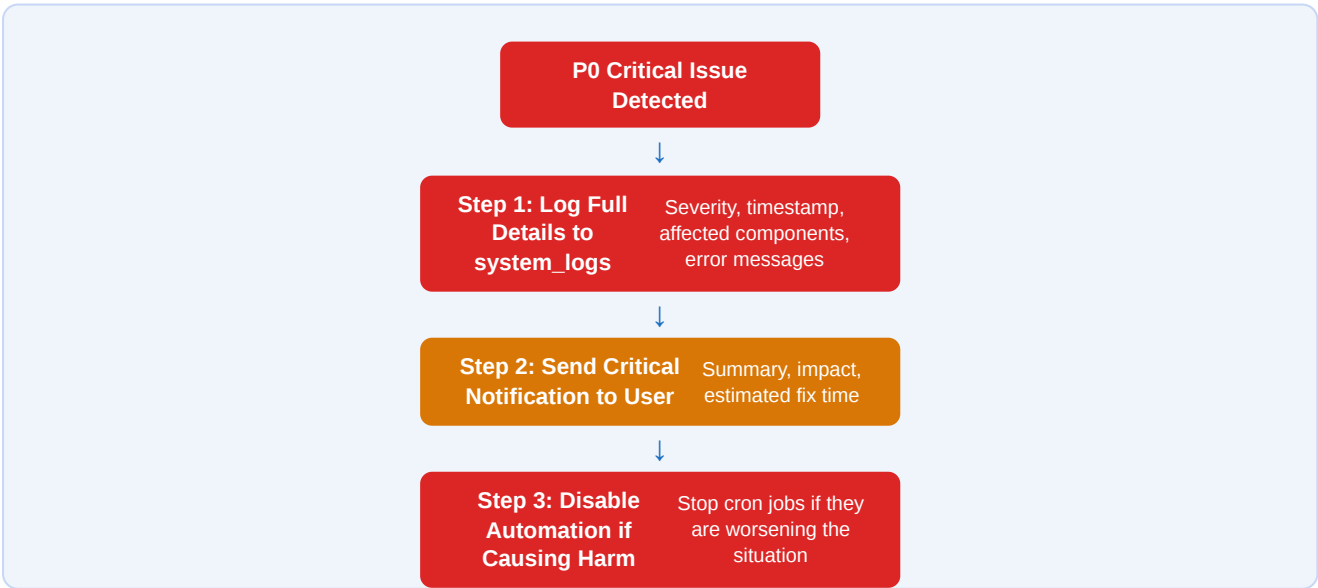
Contact User (Approval First)	Decide Autonomously (Notify After)	Flag for Awareness (No Action)
Budget impact >10% of monthly allocation	Bug fixes — always fix immediately	Approaching 80% quota/budget threshold
Architectural changes affecting scalability	Implementation details and coding approach	Unexpected but non-breaking behavior
Strategic direction changes	File structure and code organization	Performance changes (faster/slower)
New city launch timing	Minor UX improvements within spec	Growth Lab monthly summary
Tier upgrade decision (Creatomate or YouTube)	Optimization based on Growth Lab data	New Growth Lab experiment results ready
Multiple OAuth projects needed	Logging enhancements	Content performance milestones
Feature requests that change project scope	Error handling improvements	Subscriber count milestones

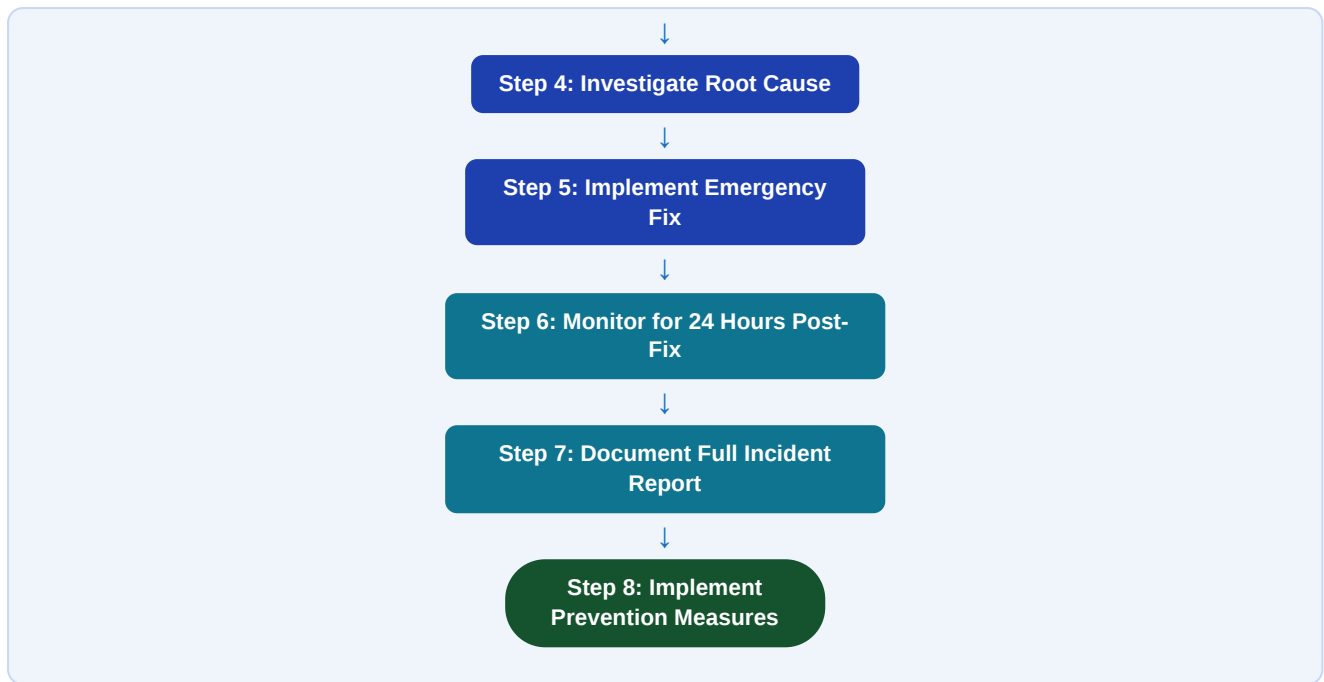
11.2

Emergency Escalation (P0 Critical Issues)

P0 Trigger Conditions — Escalate Immediately

Zero content posting for 6+ hours • YouTube API completely down • Creatomate API completely down • Database corruption or data loss • Security breach or unauthorized access detected





SECTION 12

Appendix

Glossary, key metrics reference, contacts, and revision history

A

Glossary of Terms

Project A YouTube OAuth application handling Shorts uploads. 10,000 unit daily quota, used primarily for 6 shorts/day across both cities.	Project B YouTube OAuth application handling longer-form Recap uploads. 10,000 unit daily quota. Used for Daily, Weekly, and Monthly Recaps.
Growth Lab A/B testing system that experiments with content variations (titles, thumbnails, format) and automatically promotes winning variants to default.	Creatomate Video rendering service used to produce all WeatherPost videos. Charged per credit based on rendering complexity and duration.
post_history Primary database table tracking all content posted, including platform, city, content type, status, and engagement metrics.	system_logs Structured logging database table for debugging and audit trails. Records all system events, errors, warnings, and operational milestones.
Shorts 30–60 second vertical format weather update videos. Posted 3× per day (Morning, Afternoon, Evening) per city via Project A.	Daily Recap 2–3 minute comprehensive daily weather summary video. Posted at 8 PM EDT per city via Project B.
Weekly Recap 2–3 minute weekly weather summary posted every Sunday at 2 PM EDT. Aggregates 7 days of weather data per city.	Monthly Recap 3–4 minute comprehensive monthly weather review. Triggers at 8 PM EDT on the last day of each month.
SkyBrief The WeatherPost operations dashboard where content scheduling, settings, Growth Lab experiments, and notifications are managed.	OAuth Project A Google Cloud project providing YouTube Data API access credentials. Each project has its own 10,000 unit daily quota.

B

Key Metrics Reference

YouTube API Costs (Units) Video upload1,600 units Community post50 units Daily quota10,000 / project 6 shorts/day~9,600 units 2 recaps/day~3,200 units	Creatomate Target Costs Short video≤ 5 credits Daily Recap≤ 35 credits Weekly Recap≤ 75 credits Monthly Recap≤ 120 credits Growth tier10,000 / month	Growth Milestones Monetization subs1,000 subs Watch hours req.4,000 hours Cities 1–3 target12 months Cities 4+ target9 months 10-city targetYear 2
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Content Volume Summary (2 Cities)

Content Type	Daily	Weekly	Monthly	Annual
Shorts	6	42	~180	~2,160
Daily Recaps	2	14	~60	~730
Weekly Recaps	—	2	~8	~104
Monthly Recaps	—	—	2	24
Total Posts	8	58	~250	~3,018

C Contact Information

Role	Name / Entity	Details
Project Owner	Bruce Love	Strategic decisions, budget approvals, new city launches
Developer / AI System	Lovable AI	Feature implementation, bug fixes, optimization, autonomous operations
Project Start	May 21, 2026	Initial launch with Orlando + Gainesville
Active Cities	Orlando, Gainesville	Full automation active; Growth Lab running
Next Planned City	Miami, FL	Target: July 2026 (pending 30-day stability confirmation)
Document Owner	Lovable AI	Maintains and updates this document as operations evolve

D Document Revision History

Version	Date	Author	Changes
v1.0	June 1, 2026	Lovable AI	Initial release. Covers: Daily/Weekly/Monthly operations, Bug Resolution (P0–P3), Feature Implementation (4-phase), Growth Lab lifecycle, City Launch checklist, Quota + Credit Management, Decision Trees (4), Escalation paths, and full Appendix.
v1.1	TBD	—	Reserved for Miami launch updates and any process changes from Growth Lab learnings.
v2.0	TBD	—	Major revision planned for 10-city scale milestone. Will incorporate learned operational patterns and updated automation architecture.

Document Maintenance Policy

This document should be updated whenever: (1) a new city is launched, (2) a new content type is added, (3) a significant operational change occurs, (4) Growth Lab produces a finding that changes standard practice, or (5)

a P0/P1 bug reveals a gap in the workflow documentation. The version number increments with each update; minor fixes use x.x notation, major revisions use x.0.

END OF DOCUMENT

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